

5.7 RATE OF CLIMB

The rate-of-climb charts are presented in two groups; the first group for operations with all engines operating (AEO, Fig. 5-11 and 5-12) and the second group for operations with one engine inoperative (OEI, Fig. 5-13 and 5-14). Climb rates are functions of power, pressure altitude, outside air temperature, and gross mass.

NOTE • Chart data are based on the clean configuration.

- Chart derived rate-of-climb values are established using the best rate-of-climb speed (V_Y).
- For AEO operations requiring engine anti-icing ON, reduce the chart derived rate-of-climb value by 300 fpm.
- For OEI operations requiring engine anti-icing ON, reduce chart derived rate-of-climb value by 200 fpm.

EFFECTIVITY Variants CB-4, CDN-B-4, CBS-4, and CDN-BS-4 when GM is between 2400 and 2500 kg

AEO Rate of Climb (Fig. 5-11 and 5-12)

1. Reduce the known gross mass by 100 kg.
2. Determine the rate-of-climb value using this corrected gross mass.
3. To obtain final rate-of-climb value for AEO operations, reduce the chart derived rate-of-climb value by 200 fpm.

OEI Rate of Climb (Fig. 5-13 and 5-14)

To obtain final rate-of-climb value for OEI operations, reduce chart derived rate-of-climb value by 100 fpm.

EFFECTIVITY ALL

EXAMPLE: (see Fig. 5-11)

Determine: **Rate of climb**

Known:	Pressure altitude	11000 ft
	OAT	20 °C
	Gross mass	1820 kg

Solution:

1. Enter chart at known pressure altitude (11000 ft).
2. Move up to known OAT (20 °C).
3. Move right to reference line and then down following curve of guide lines until intercepting known gross mass (1820 kg).
4. From this intercept point, move right and read rate of climb = 1420 fpm.

RATE OF CLIMB
2 X ALLISON 250-C20B

AEO – TAKEOFF POWER (810 °C TOT, 86% TORQUE)

ANTI-ICING – OFF

BEST RATE-OF-CLIMB SPEED – 60 KIAS

BLEED AIR CONSUMERS – OFF

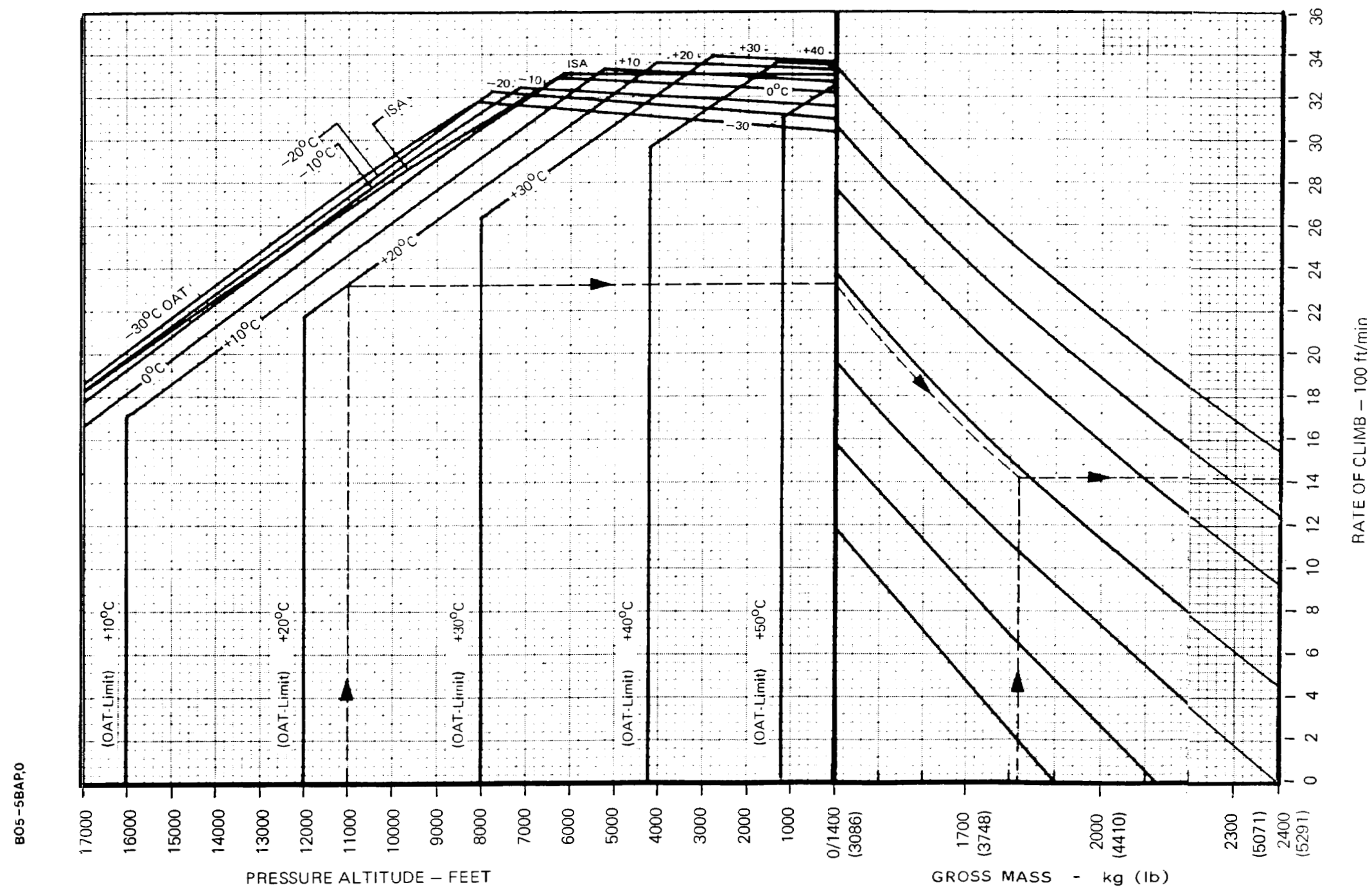


Fig. 5-11 Rate of Climb - AEO, TOP (Sheet 1 of 2)

RATE OF CLIMB

EXTREME LOW TEMPERATURES (-30 TO -45 °C)
2 X ALLISON 250-C20B

AEO – TAKEOFF POWER (810 °C TOT, 86% TORQUE)

ANTI-ICING – OFF

BEST RATE-OF-CLIMB SPEED – 60 KIAS

BLEED AIR CONSUMERS – OFF

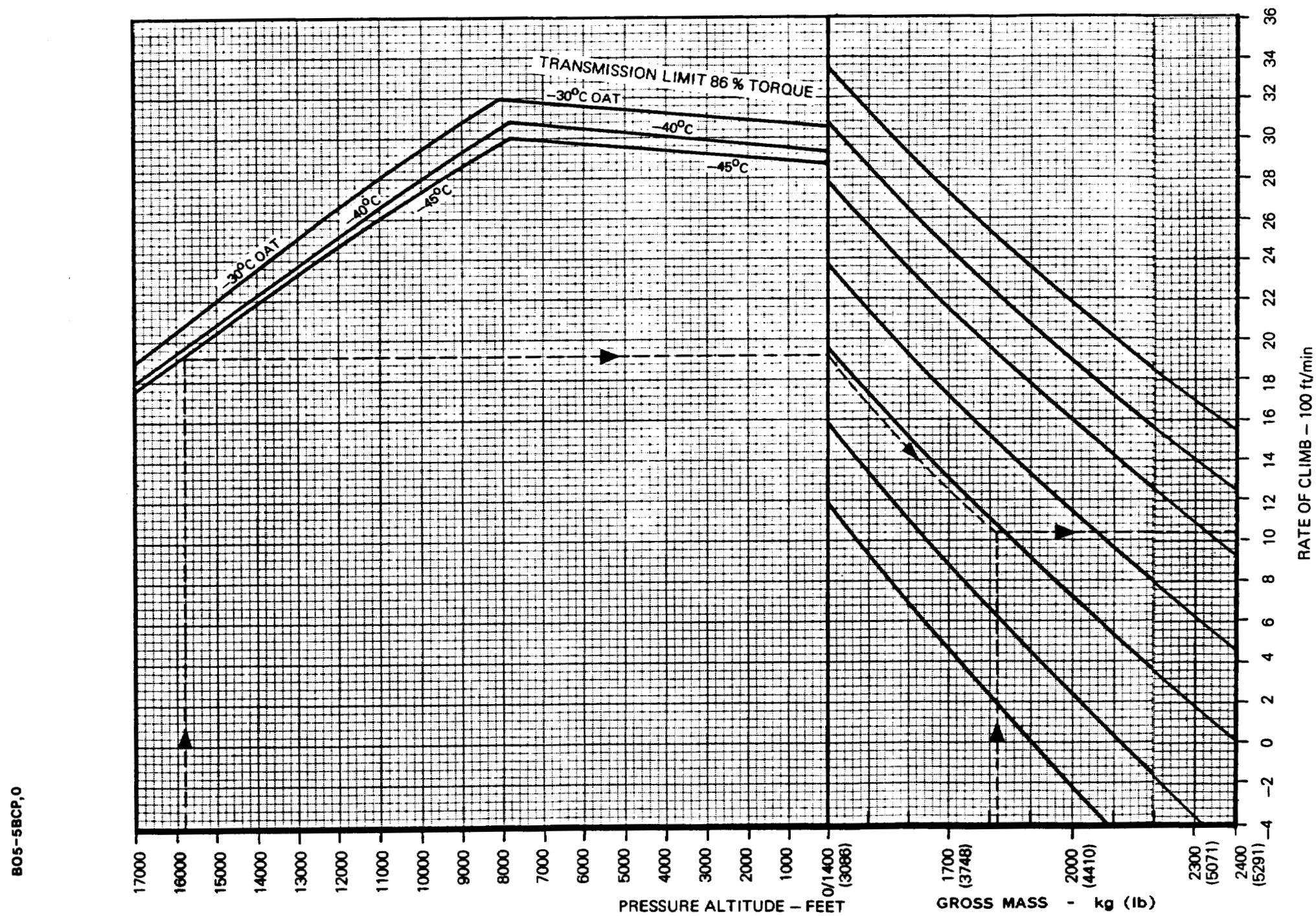


Fig. 5-11 Rate of Climb - AEO, TOP (Sheet 2 of 2)

RATE OF CLIMB

2 X ALLISON 250-C20B

AEO – MAX CONTINUOUS POWER (779 °C TOT, 86% TORQUE)

ANTI-ICING – OFF

BEST RATE-OF-CLIMB SPEED – 60 KIAS

BLEED AIR CONSUMERS – OFF

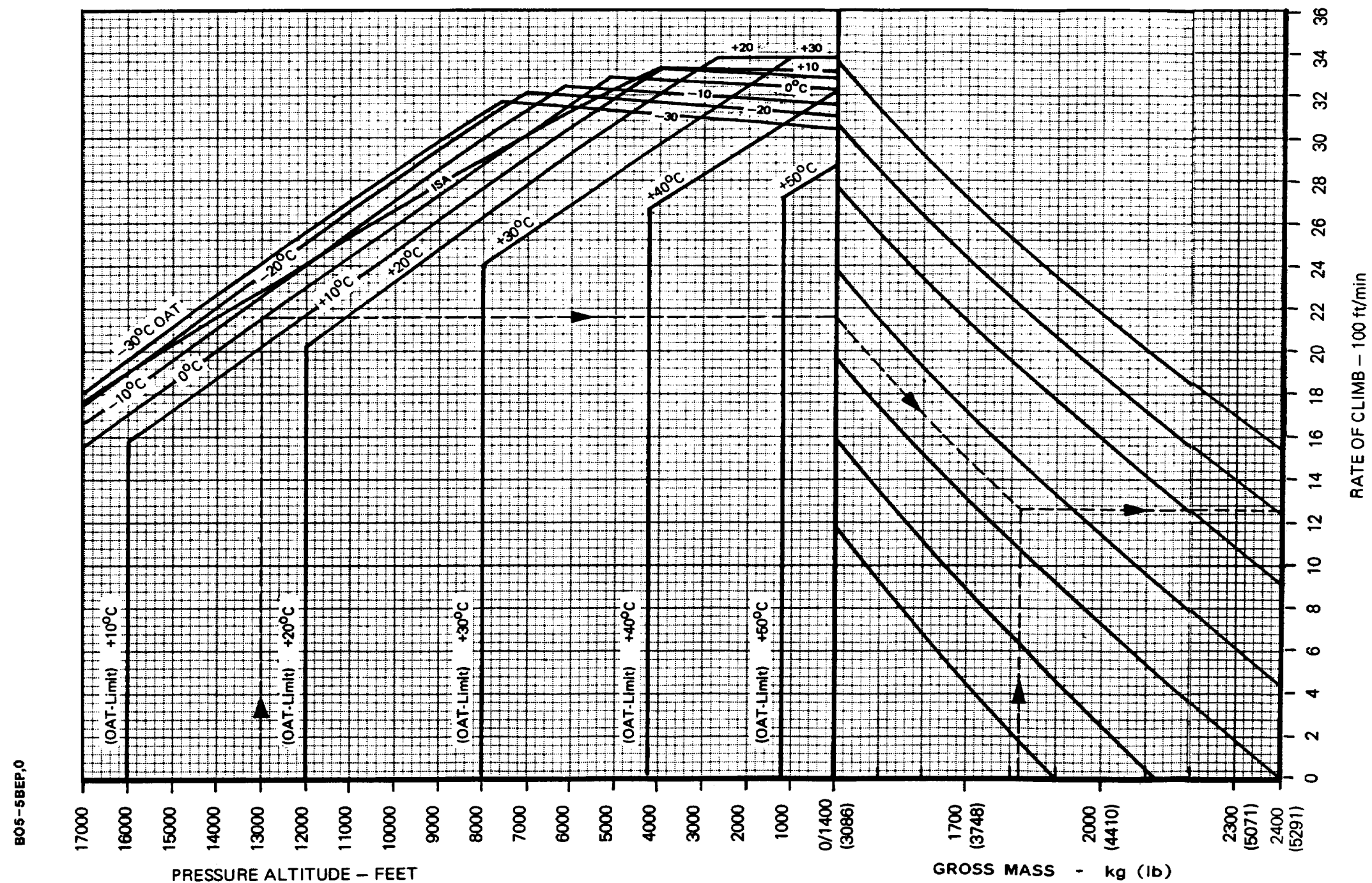


Fig. 5-12 Rate of Climb - AEO, MCP (Sheet 1 of 2)

RATE OF CLIMB

EXTREME LOW TEMPERATURES (-30 TO -45 °C)
2 X ALLISON 250-C20B

AEO – MAX CONTINUOUS POWER (779 °C TOT, 86% TORQUE)

ANTI-ICING – OFF

BEST RATE-OF-CLIMB SPEED – 60 KIAS

BLEED AIR CONSUMERS – OFF

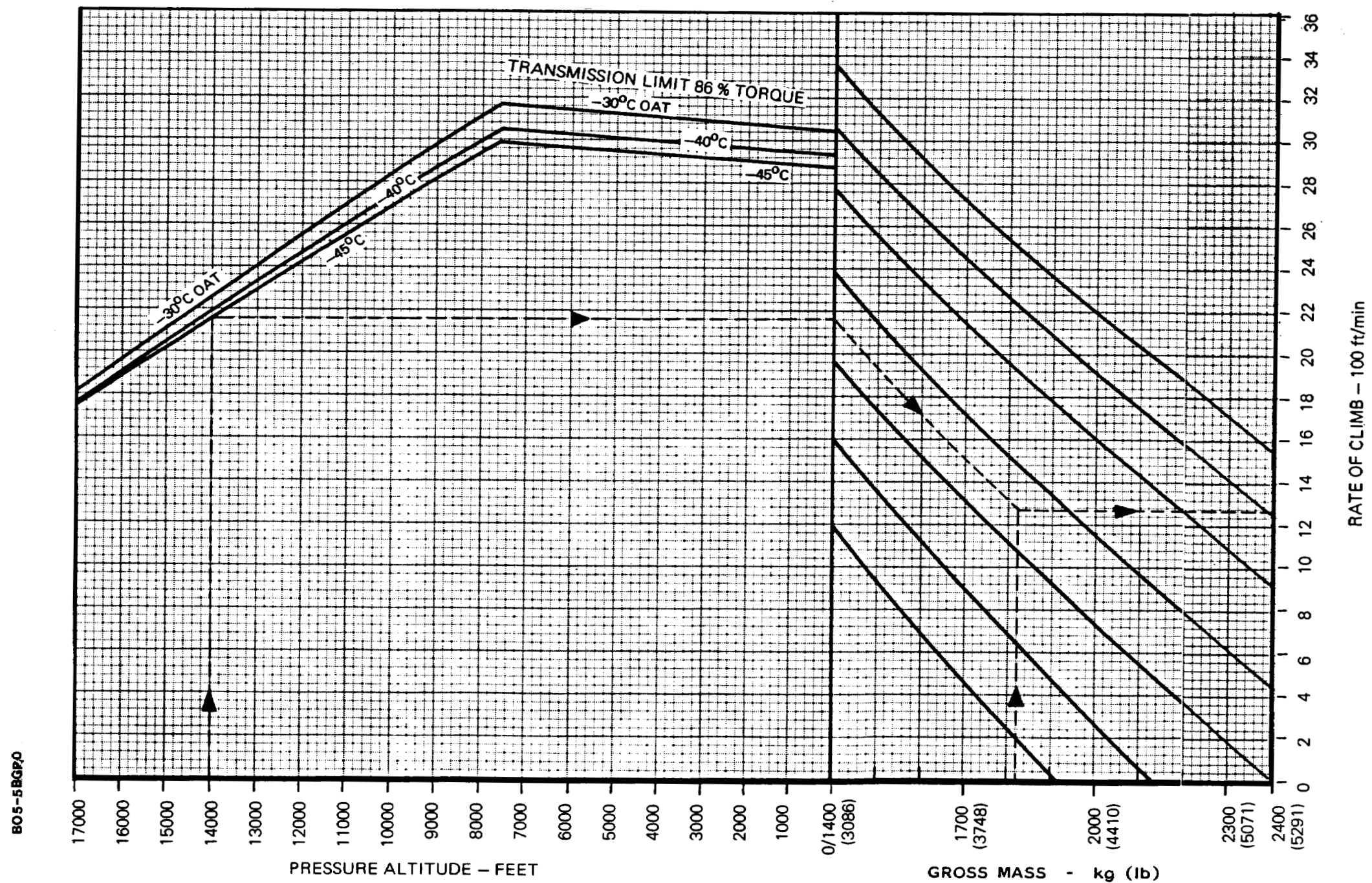


Fig. 5-12 Rate of Climb - AEO, MCP (Sheet 2 of 2)

RATE OF CLIMB 1 X ALLISON 250-C20B



OEI – EMERGENCY POWER (810 °C TOT, 95% TORQUE)

ANTI-ICING – OFF

BEST RATE-OF-CLIMB SPEED – 60 KIAS

BLEED AIR CONSUMERS – OFF

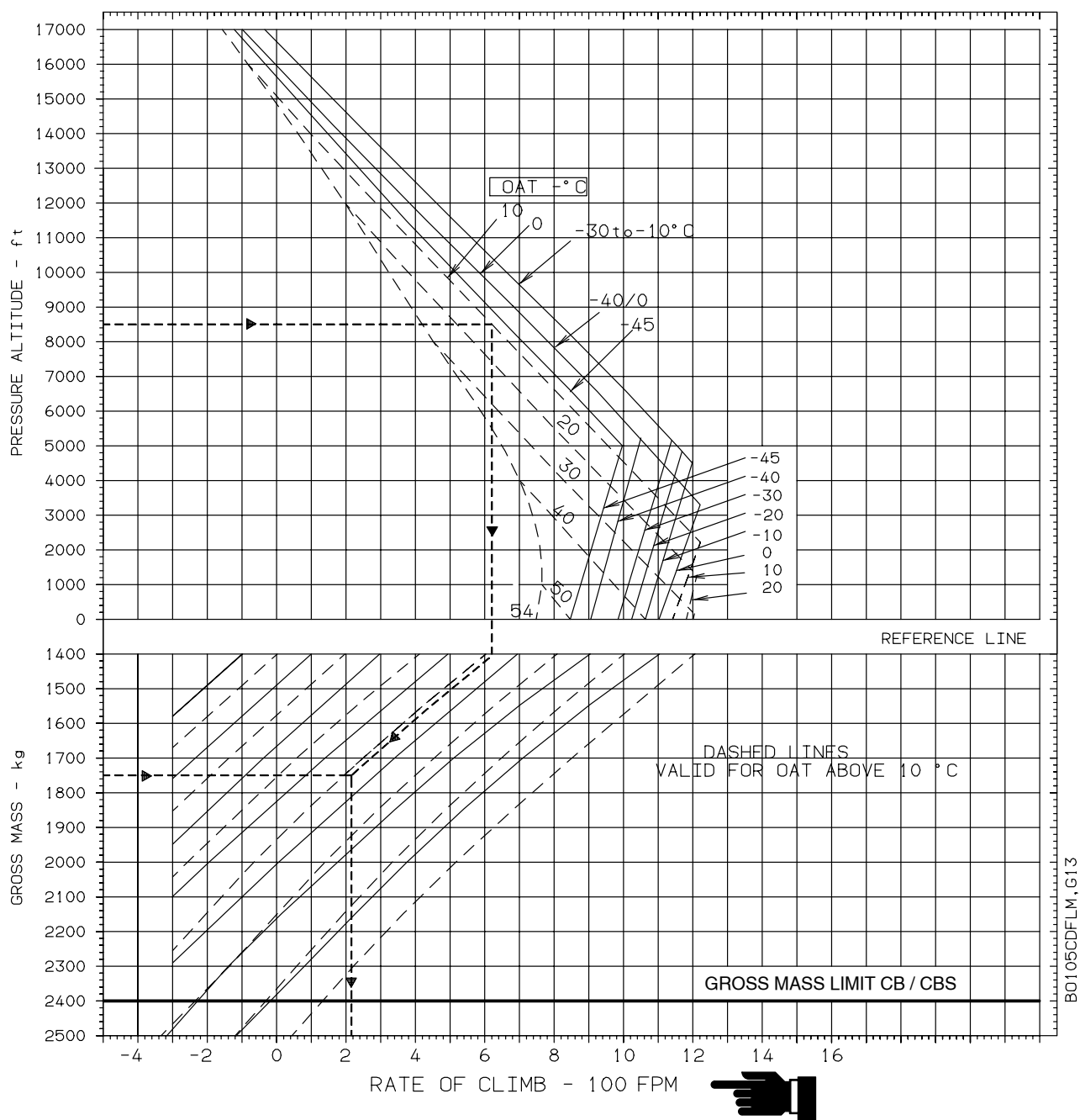
NOTE WITH ENGINE ANTI-ICING ON THE RATE OF CLIMB IS REDUCED BY 200 FPM.

Fig. 5-13 Rate of Climb - OEI, EPWR

RATE OF CLIMB 1 X ALLISON 250-C20B



OEI – MAX CONTINUOUS POWER (779 °C TOT, 95% TORQUE)

ANTI-ICING – OFF

BEST RATE-OF-CLIMB SPEED – 60 KIAS

BLEED AIR CONSUMERS – OFF

NOTE WITH ENGINE ANTI-ICING ON THE RATE OF CLIMB IS REDUCED BY 200 FPM.

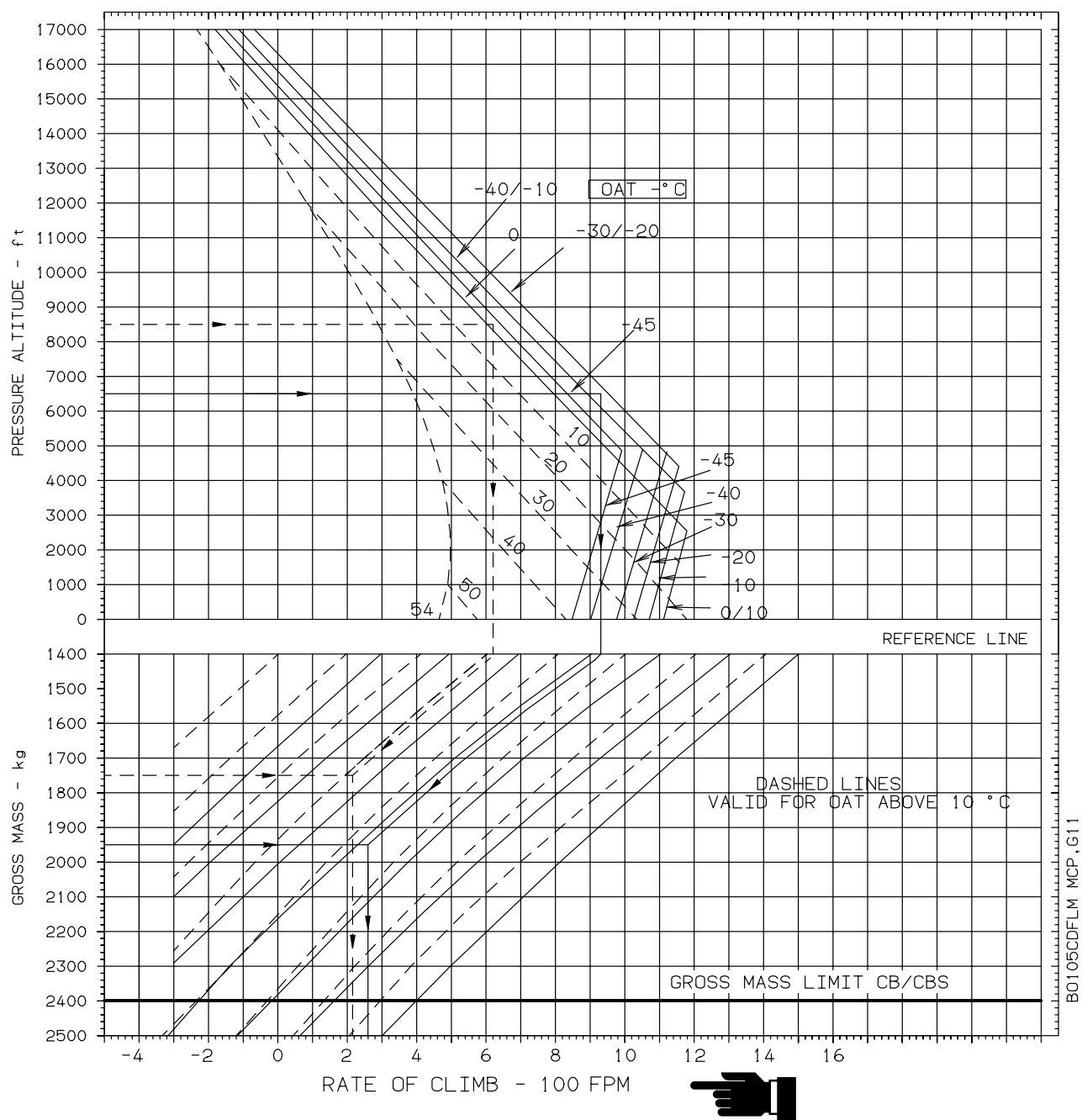


Fig. 5-14 Rate of Climb - OEI, MCP

LBA APPROVED

Rev. 2